



University
of Glasgow

Level 1

OKD4 Cluster

Getting started with Launcher

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WORLD
CHANGING
GLASGOW

A WORLD
TOP 100
UNIVERSITY

How Do I Use it?

What you **need to do** will define **what you need to learn** to use the cluster, you can think of it as having **4 levels**, where you incrementally need to understand more of how the cluster works as you need to undertake more complex work

You are Here

your account, resources and storage



Level 1

- Cluster users who just need to get up and running with Python/Jupyter or need remote execution with VSCode are recommended to use the Launcher tool



Level 2

- For scenarios where you need to deploy a wider range of existing services, you will need to learn the basics of Containers and Kubernetes. This will allow you to deploy pre-built container solutions from platforms like Dockerhub or more complex services via OperatorHub.



Level 3

- For running batch experiments (queuing many containers to run concurrently) you will need to learn how to interact with the cluster via the command line tool (oc)



Level 4

- If you need to create a new development environment from first principles or custom applications/demos, then you will need to learn how to produce new containers via Builds, and it is recommended to learn how to create continuous integration pipelines.

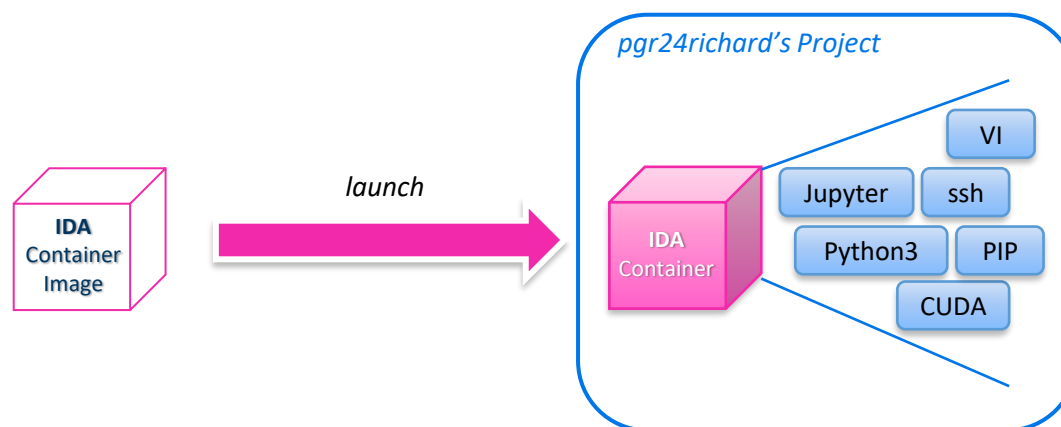
The Standard Container

The majority of our users only need a basic working **Unix container** to do their research

- This is because once you have a running container, you **can install more software onto it as needed**, customising it for your use-cases
- In effect a container can act like a **machine you have root/admin permissions on**

Hence, **we provide a pre-built standard IDA container (image)** that is tested and **compatible with our cluster** such that you don't need to build your own

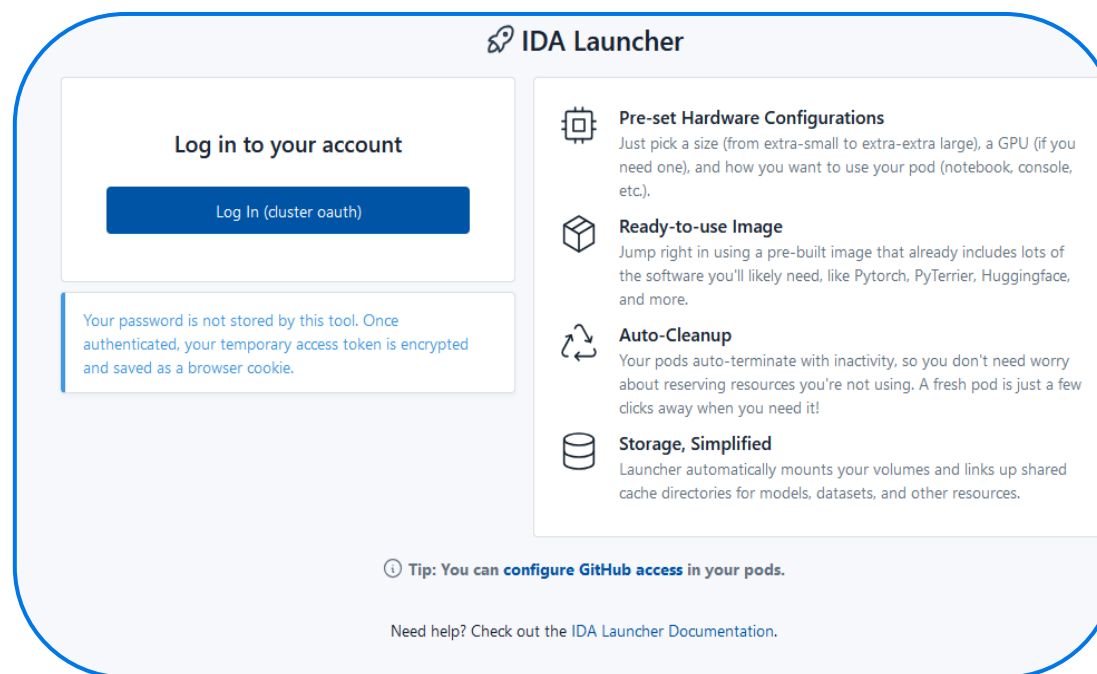
- This container image comes **pre-installed** with many **commonly used libraries**
- When you want to do work, you **launch** a container based on this **image** into your project space



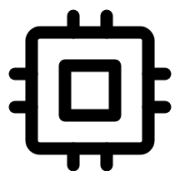
Making Launching Easier

In the past you would need a basic understanding of our cluster management system to launch the container, however we have developed an application to make this just a few clicks:

<https://launcher.apps.os.dcs.gla.ac.uk/>



Launcher Features



Pre-set Hardware Configurations

Just pick a size (from extra-small to extra-extra large), a GPU (if you need one), and how you want to use your pod (notebook, console, etc.).



Ready-to-use Image

Jump right in using a pre-built image that already includes lots of the software you'll likely need, like Pytorch, PyTerrier, Huggingface, and more.



Auto-Cleanup

Your pods auto-terminate with inactivity, so you don't need worry about reserving resources you're not using. A fresh pod is just a few clicks away when you need it!

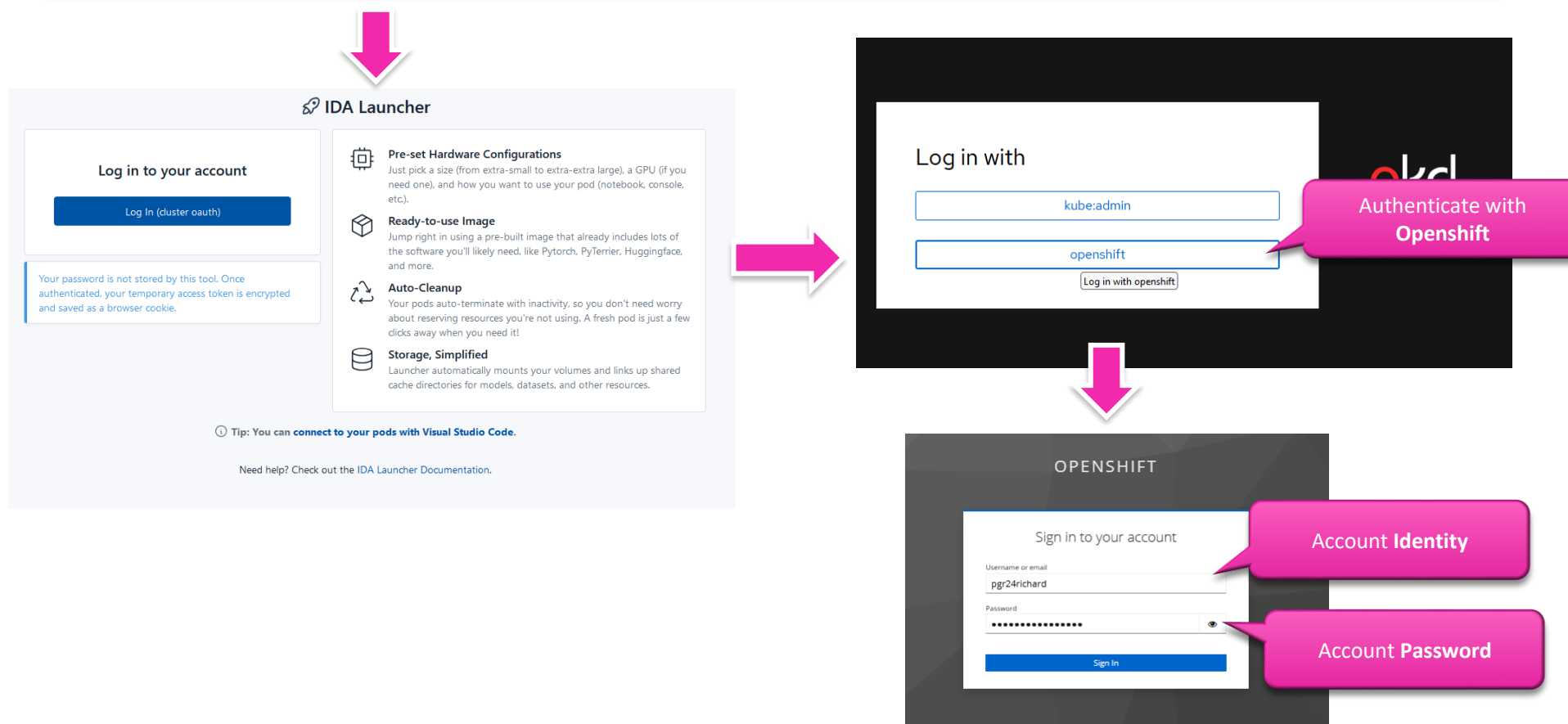


Simplified Storage

Launcher automatically mounts your volumes and links up shared cache directories for models, datasets, and other resources.

Step 1: Log-into Launcher

<https://launcher.apps.os.dcs.gla.ac.uk/>



➤ The first time you log in you will also get asked by Launcher for your email address for [alerts](#)

Launcher Home Screen

- This is the home screen for launcher, there are some important controls here, so let's go over them...

 IDA Launcher

richardm

Quick Notebook

Launch a Pod

Log Out

USER

richardm

STORAGE

► Volume Claims (click to expand)

► Global Environment Variables (click to expand)

PROJECT / NAMESPACE

richardm

SECURITY

► Pod SSH Key (click to expand)

► SSH Access (click to expand)

GPUS RESERVED

0 / 1

Number of GPUs requested / limit

Logged in User

Target Project

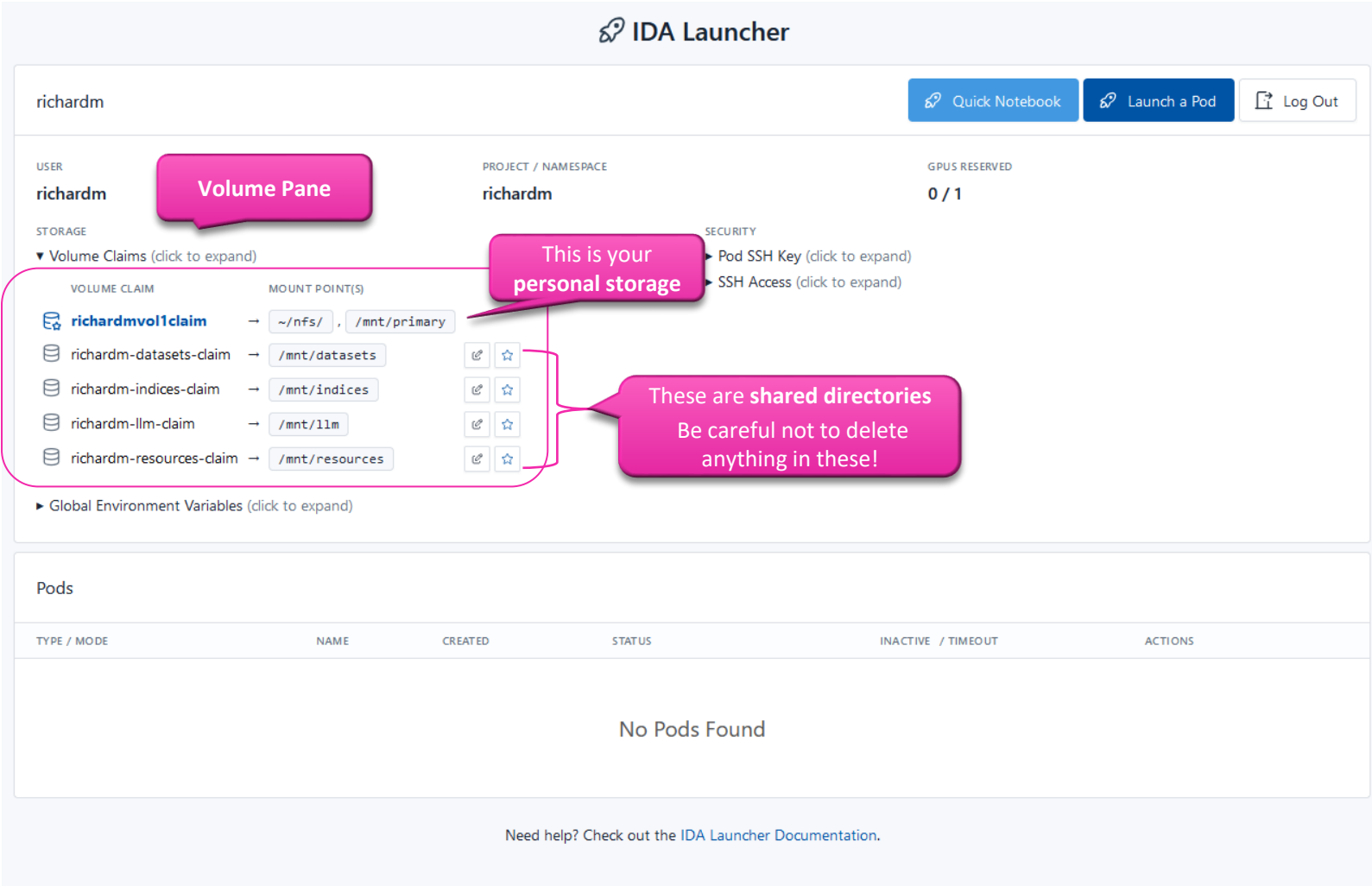
Pods

TYPE / MODE	NAME	CREATED	STATUS	INACTIVE / TIMEOUT	ACTIONS
No Pods Found					

Need help? Check out the [IDA Launcher Documentation](#).

Launcher Home Screen

- The volume pane (when expanded) shows you **what persistent storage** will be mounted onto your container and **where it will appear**



The screenshot shows the IDA Launcher interface for user 'richardm'. The 'Volume Pane' is expanded, showing a table of volume claims. A callout points to the 'richardmvol1claim' as 'personal storage'. Another callout points to the other claims as 'shared directories' with a warning not to delete anything. The 'Pods' section at the bottom shows 'No Pods Found'.

IDA Launcher

richardm [Quick Notebook](#) [Launch a Pod](#) [Log Out](#)

USER: richardm PROJECT / NAMESPACE: richardm GPUS RESERVED: 0 / 1

STORAGE: [Volume Claims \(click to expand\)](#) SECURITY: [Pod SSH Key \(click to expand\)](#) [SSH Access \(click to expand\)](#)

VOLUME CLAIM	MOUNT POINT(S)	
 richardmvol1claim	→ ~/nfs/ , /mnt/primary	 
 richardm-datasets-claim	→ /mnt/datasets	 
 richardm-indices-claim	→ /mnt/indices	 
 richardm-llm-claim	→ /mnt/llm	 
 richardm-resources-claim	→ /mnt/resources	 

► [Global Environment Variables \(click to expand\)](#)

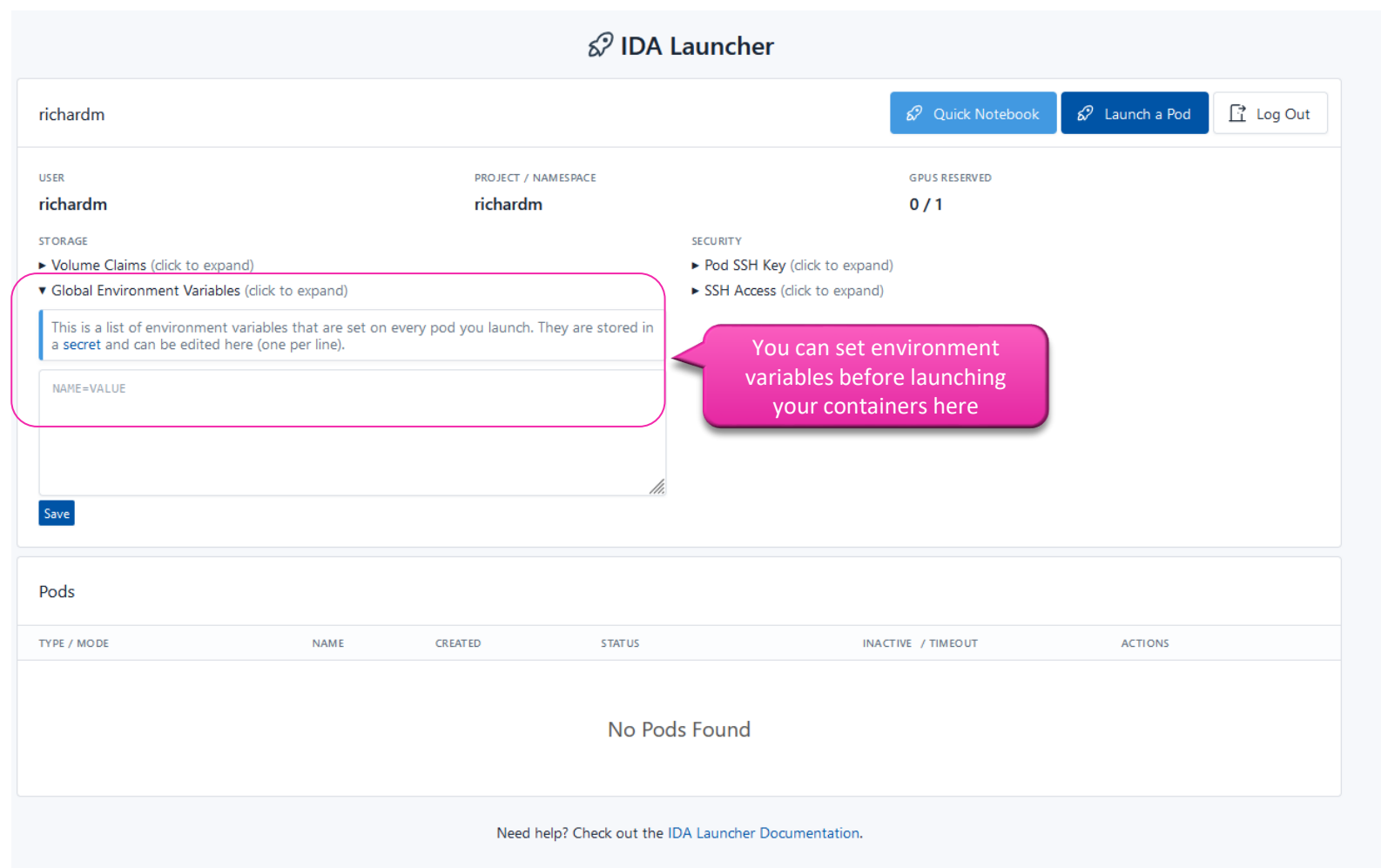
Pods

TYPE / MODE	NAME	CREATED	STATUS	INACTIVE / TIMEOUT	ACTIONS
No Pods Found					

Need help? Check out the [IDA Launcher Documentation](#).

Launcher Home Screen

- The Global Environment Variables pane allows you to set **environment variables** before launching a pod



The screenshot shows the IDA Launcher interface. At the top, the user 'richardm' is logged in. Below this, there are sections for USER, PROJECT / NAMESPACE, and GPUS RESERVED. The main content area is divided into STORAGE and SECURITY sections. The STORAGE section includes 'Volume Claims' and 'Global Environment Variables'. The 'Global Environment Variables' section is highlighted with a pink box and contains a text area for setting variables, with a pink callout bubble stating: 'You can set environment variables before launching your containers here'. The SECURITY section includes 'Pod SSH Key' and 'SSH Access'. Below these sections is a 'Pods' table, which is currently empty, showing 'No Pods Found'. At the bottom, there is a link to the IDA Launcher Documentation.

IDA Launcher

richardm

Quick Notebook Launch a Pod Log Out

USER: richardm PROJECT / NAMESPACE: richardm GPUS RESERVED: 0 / 1

STORAGE

- ▶ Volume Claims (click to expand)
- ▼ Global Environment Variables (click to expand)

This is a list of environment variables that are set on every pod you launch. They are stored in a **secret** and can be edited here (one per line).

NAME=VALUE

Save

SECURITY

- ▶ Pod SSH Key (click to expand)
- ▶ SSH Access (click to expand)

You can set environment variables before launching your containers here

Pods

TYPE / MODE	NAME	CREATED	STATUS	INACTIVE / TIMEOUT	ACTIONS
No Pods Found					

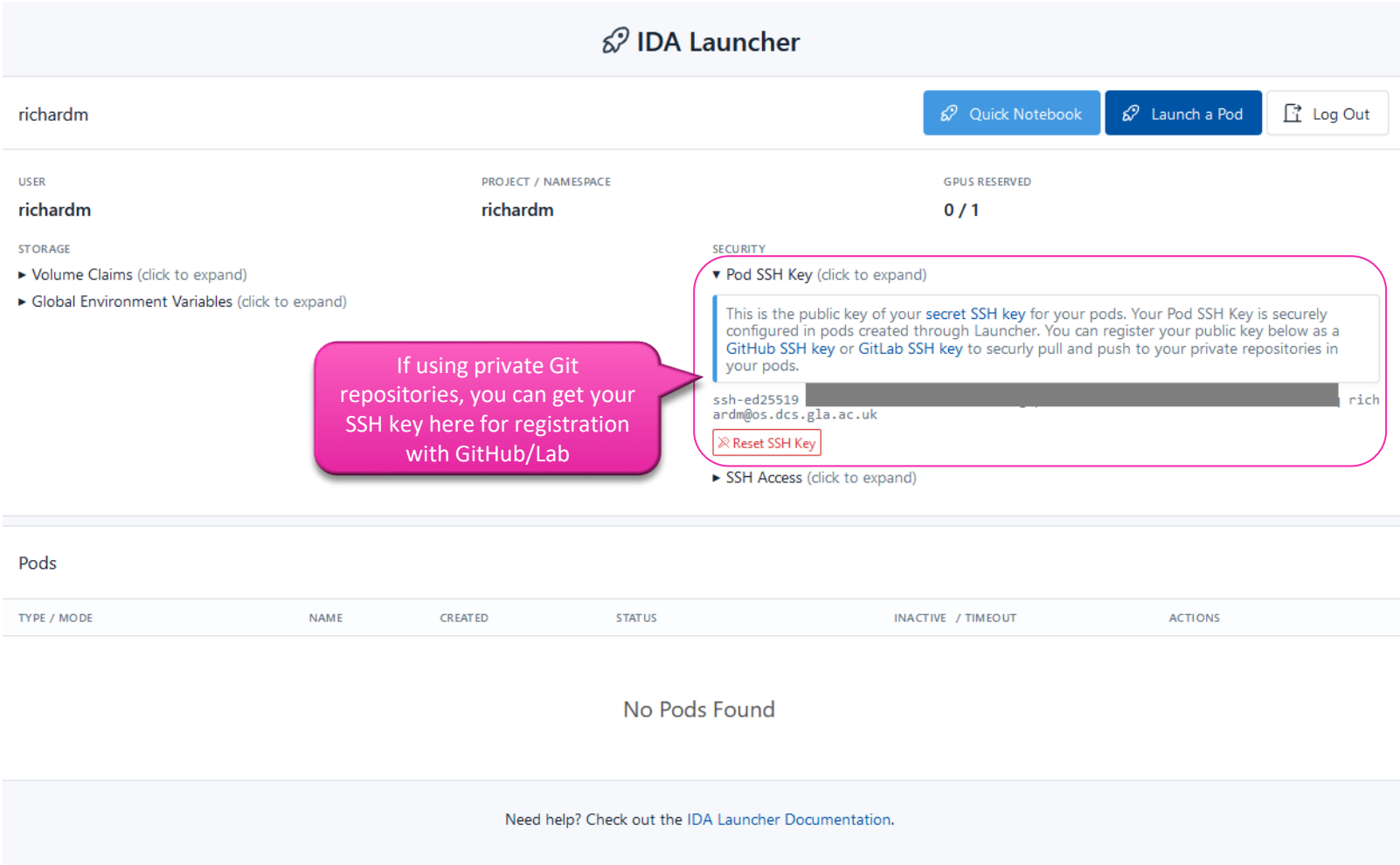
Need help? Check out the [IDA Launcher Documentation](#).

Launcher Home Screen

- Some services only accept connections from known hosts, where you have registered that host by providing an **SSH key**

➤ Your containers can act like a host, and you can get the SSH key from here

- This is useful if using **GitHub** or **GitLab**
 - Keep your repositories private by default!



IDA Launcher

richardm

Quick Notebook Launch a Pod Log Out

USER: richardm PROJECT / NAMESPACE: richardm GPUS RESERVED: 0 / 1

STORAGE

- ▶ Volume Claims (click to expand)
- ▶ Global Environment Variables (click to expand)

SECURITY

▼ Pod SSH Key (click to expand)

This is the public key of your **secret SSH key** for your pods. Your Pod SSH Key is securely configured in pods created through Launcher. You can register your public key below as a **GitHub SSH key** or **GitLab SSH key** to securely pull and push to your private repositories in your pods.

ssh-ed25519 ardm@os.dcs.gla.ac.uk rich

Reset SSH Key

SSH Access (click to expand)

Pods

TYPE / MODE	NAME	CREATED	STATUS	INACTIVE / TIMEOUT	ACTIONS
No Pods Found					

Need help? Check out the [IDA Launcher Documentation](#).

University of Glasgow

- You can give your container your local public key here, so it knows you when you try to make an SSH connection

richardm

Quick Notebook

Launch a Pod

Log Out

USER

richardm

PROJECT / NAMESPACE

richardm

GPU'S RESERVED

0 / 1

STORAGE

► Volume Claims (click to expand)

► Global Environment Variables (click to expand)

SECURITY

► Pod SSH Key (click to expand)

▼ SSH Access (click to expand)

1 Generate your Personal SSH Key

If you do not yet have an SSH key, generate one. GitHub has a good guide [here](#).

2 Configure SSH

Run the following commands, which route and verify your cluster SSH connections.

```
echo -e "\nHost *.os\n\tProxyCommand cat <(echo -e \"CONNECT %h:%p HTTP/1.0\\\\\\r\\\\\\nHost: ssh.launcher.apps.os.dcs.gla.ac.uk\\\\\\r\\\\\\n\" - | nc ssh.launcher.apps.os.dcs.gla.ac.uk 80)\n\tuser root\n\" >> ~/.ssh/config
echo \"*.*.os ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIAIgcPa8do9B3zFF1AUJ25K3oeZVPeUI53tS27ziITUq richardm@os.dcs.gla.ac.uk\" >> ~/.ssh/known_hosts
```

3 Authorise your Personal SSH Public Key

Run the following command, copy/paste the result below, and save. (This lets the pods verify your identity.)

```
cat ~/.ssh/id_*.pub
```

Save

4 Finish the steps above

You can give your container your local public key here, so it knows you when you try to make an SSH connection

Pods

TYPE / MODE	NAME	CREATED	STATUS	INACTIVE / TIMEOUT	ACTIONS
No Pods Found					

Step 2: Prepare to Launch

- With that covered, lets
launch a pod

 IDA Launcher

richardm

[Quick Notebook](#) [Launch a Pod](#) [Log Out](#)

USER

richardm

PROJECT / NAMESPACE

richardm

GPUS RESERVED

0 / 1

STORAGE

[▶ Volume Claims \(click to expand\)](#)
[▶ Global Environment Variables \(click to expand\)](#)

SECURITY

[▶ Pod SSH Key \(click to expand\)](#)
[▶ SSH Access \(click to expand\)](#)

Pods

TYPE / MODE	NAME	CREATED	STATUS	INACTIVE / TIMEOUT	ACTIONS
No Pods Found					

Need help? Check out the [IDA Launcher Documentation](#).

Click HERE!

Step 3: Configure the Container

- This is the **launch screen**, here you specify container:

- Resource Request
- Container Start Mode
- Automatic Clean-up

You can configure the container name here, otherwise it will be automatically generated

IDA Launcher

POD NAME sleepy-alpaca-51	USERNAME richardm	NAMESPACE richardm	PRIMARY VOLUME richardmvol1claim
------------------------------	----------------------	-----------------------	-------------------------------------

Hardware Select the hardware specification of the pod to launch

	Size: X-Small 1 CPU · 1GB RAM	Size: Small 2 CPU · 8GB RAM	Size: Medium 2 CPU · 16GB RAM	Size: Large 4 CPU · 32GB RAM	Size: X-Large 8 CPU · 64GB RAM	Size: X-X-Large 16 CPU · 128GB RAM
GPU: None	cpu-xs	cpu-sm	cpu-md	cpu-lg	cpu-xl	cpu-xxl
GPU: TITAN 24GB vRAM	titan-xs	titan-sm	titan-md	titan-lg	titan-xl	titan-xxl
GPU: 3090 24GB vRAM	3090-xs	3090-sm	3090-md	3090-lg	3090-xl	3090-xxl
GPU: 4090 24GB vRAM	4090-xs	4090-sm	4090-md	4090-lg	4090-xl	4090-xxl
GPU: A6000 48GB vRAM	a6000-xs	a6000-sm	a6000-md	a6000-lg	a6000-xl	a6000-xxl

Mode How do you want to interact with this pod?

Console
Use the command line

Notebook
Use a Jupyter notebook

Script
Run the specified script

Timeout Your pod will be deleted after prolonged inactivity

10 minutes
Best for compute pods

2 hours
Best for interactive pods

► Advanced Options (click to expand)

[Launch!](#)

Need help? Check out the [IDA Launcher Documentation](#).

Step 3: Configure the Container

- Remember our earlier discussion on requesting resources?
- To make things simpler we provide a grid of options for you of **standardized sizes**

IDA Launcher

POD NAME: sleepy-alpaca-51 USERNAME: richardm NAMESPACE: richardm PRIMARY VOLUME: richardmvol1claim

Hardware Select the hardware specification of the pod to launch

	Size: X-Small 1 CPU · 1GB RAM	Size: Small 2 CPU · 8GB RAM	Size: Medium 2 CPU · 16GB RAM	Size: Large 4 CPU · 32GB RAM	Size: X-Large 8 CPU · 64GB RAM	Size: X-X-Large 16 CPU · 128GB RAM
GPU: None 24GB vRAM	cpu-xs	cpu-sm	cpu-md	cpu-lg	cpu-xl	cpu-xxl
GPU: TITAN 24GB vRAM	titan-xs	titan-sm	titan-md	titan-lg	titan-xl	titan-xxl
GPU: 3090 24GB vRAM	3090-xs	3090-sm	3090-md	3090-lg	3090-xl	3090-xxl
GPU: 4090 24GB vRAM	4090-xs	4090-sm	4090-md	4090-lg	4090-xl	4090-xxl
GPU: A6000 48GB vRAM	a6000-xs	a6000-sm	a6000-md	a6000-lg	a6000-xl	a6000-xxl

Mode How do you want to interact with this pod?

Console
Use the command line

Notebook
Use a Jupyter notebook

Script
Run the specified script

Advanced Options (click to expand)

Launch!

Need help? Check out the IDA Launcher Documentation.

Horizontal is CPU and RAM

Vertical is GPU

The little number here specifies how many containers of that size the cluster has space to support (at that moment)

If its greyed out, there is not enough capacity

Step 3: Configure the Container

Launcher can currently support three launch modes:

- **Console**: Start the container and connect to its (bash) console
- **Notebook**: Start the container running a Jupyter Lab server
- **Script**: Start the container and execute a script you have in your personal storage folder

IDA Launcher

POD NAME
sleepy-alpaca-51

USERNAME
richardm

NAMESPACE
richardm

PRIMARY VOLUME
richardmvoltclaim

Hardware

Select the hardware specification of the pod to launch

	Size: X-Small 1 CPU · 1GB RAM	Size: Small 2 CPU · 8GB RAM	Size: Medium 2 CPU · 16GB RAM	Size: Large 4 CPU · 32GB RAM	Size: X-Large 8 CPU · 64GB RAM	Size: X-X-Large 16 CPU · 128GB RAM
GPU: None	cpu-xs	cpu-sm	cpu-md	cpu-lg	cpu-xl	cpu-xxl
GPU: TITAN 24GB vRAM	titan-xs	titan-sm	titan-md	titan-lg	titan-xl	titan-xxl
GPU: 3090 24GB vRAM	3090-xs	3090-sm	3090-md	3090-lg	3090-xl	3090-xxl
GPU: 4090 24GB vRAM	4090-xs	4090-sm	4090-md	4090-lg	4090-xl	4090-xxl
GPU: A6000 48GB vRAM	a6000-xs	a6000-sm	a6000-md	a6000-lg	a6000-xl	a6000-xxl

Mode

How do you want to interact with this pod?

Console
Use the command line

Notebook
Use a Jupyter notebook

Script
Run the specified script

Timeout

Your pod will be deleted after prolonged inactivity

10 minutes
Best for compute pods

2 hours
Best for interactive pods

Advanced Options (click to expand)

Launch!

Need help? Check out the IDA Launcher Documentation.

Step 3: Configure the Container

As we are sharing resources among users, we need to have a policy for releasing un-used resources

Launcher will shut-down any container after a period of consecutive (CPU) inactivity

- For undergraduates, MSci and MSc, this is 10 minutes
- Post-docs and Staff can set this higher

IDA Launcher

POD NAME
sleepy-alpaca-51

USERNAME
richardm

NAMESPACE
richardm

PRIMARY VOLUME
richardmvoltclaim

Hardware

Select the hardware specification of the pod to launch

	Size: X-Small 1 CPU · 1GB RAM	Size: Small 2 CPU · 8GB RAM	Size: Medium 2 CPU · 16GB RAM	Size: Large 4 CPU · 32GB RAM	Size: X-Large 8 CPU · 64GB RAM	Size: X-X-Large 16 CPU · 128GB RAM
GPU: None	cpu-xs	cpu-sm	cpu-md	cpu-lg	cpu-xl	cpu-xxl
GPU: TITAN 24GB vRAM	titan-xs	titan-sm	titan-md	titan-lg	titan-xl	titan-xxl
GPU: 3090 24GB vRAM	3090-xs	3090-sm	3090-md	3090-lg	3090-xl	3090-xxl
GPU: 4090 24GB vRAM	4090-xs	4090-sm	4090-md	4090-lg	4090-xl	4090-xxl
GPU: A6000 48GB vRAM	a6000-xs	a6000-sm	a6000-md	a6000-lg	a6000-xl	a6000-xxl

Mode

How do you want to interact with this pod?

Console
Use the command line

Script
Run the specified script

Notebook
Use a Jupyter notebook

Timeout

Your pod will be deleted after prolonged inactivity

10 minutes
Best for compute pods

2 hours
Best for interactive pods

Advanced Options (click to expand)

Launch!

Need help? Check out the IDA Launcher Documentation.

Step 3: Configure the Container

Once you have selected an option for all three configurables...

... click **Launch!**

IDA Launcher

POD NAME
sleepy-alpaca-51

USERNAME
richardm

NAMESPACE
richardm

PRIMARY VOLUME
richardmvolclaim

Hardware

Select the hardware specification of the pod to launch

	Size: X-Small 1 CPU · 1GB RAM	Size: Small 2 CPU · 8GB RAM	Size: Medium 2 CPU · 16GB RAM	Size: Large 4 CPU · 32GB RAM	Size: X-Large 8 CPU · 64GB RAM	Size: X-X-Large 16 CPU · 128GB RAM
GPU: None	cpu-xs	cpu-sm	cpu-md	cpu-lg	cpu-xl	cpu-xxl
GPU: TITAN 24GB vRAM	titan-xs	titan-sm	titan-md	titan-lg	titan-xl	titan-xxl
GPU: 3090 24GB vRAM	3090-xs	3090-sm	3090-md	3090-lg	3090-xl	3090-xxl
GPU: 4090 24GB vRAM	4090-xs	4090-sm	4090-md	4090-lg	4090-xl	4090-xxl
GPU: A6000 48GB vRAM	a6000-xs	a6000-sm	a6000-md	a6000-lg	a6000-xl	a6000-xxl

Mode

How do you want to interact with this pod?

Console
Use the command line

Notebook
Use a Jupyter notebook

Script
Run the specified script

Timeout

Your pod will be deleted after prolonged inactivity

10 minutes
Best for compute pods

2 hours
Best for interactive pods

Advanced Options (click to expand)

Launch!

Need help? Check out the IDA Launcher Documentation.

Step 4: Wait for Container Ready

Launcher will in the background try and **create an IDA Container** for you in a Pod

Once ready, it will give you a **link**

- This example is for a notebook server

 IDA Launcher

sleepy-alpaca-51

USER	NAMESPACE	HARDWARE TYPE	MODE
richardm	richardm	3090-md	notebook

1

Pod Created
The pod was successfully created.
[Open Pod in OpenShift Console](#)

2

Pod Scheduled
The pod was assigned to node `idagpu-09`

3

Container Started
The container has started.

4

Notebook Server Started
The notebook server has started and is ready to use.

Jupyter Notebook

[Open Notebook](#)

ⓘ

 Use this option to access JupyterLab for this pod.

Cleanup

[Delete Pod](#)

ⓘ

 Your pod will be deleted automatically after 10m of inactivity.

[Return Home](#)

[Delete Pod](#)

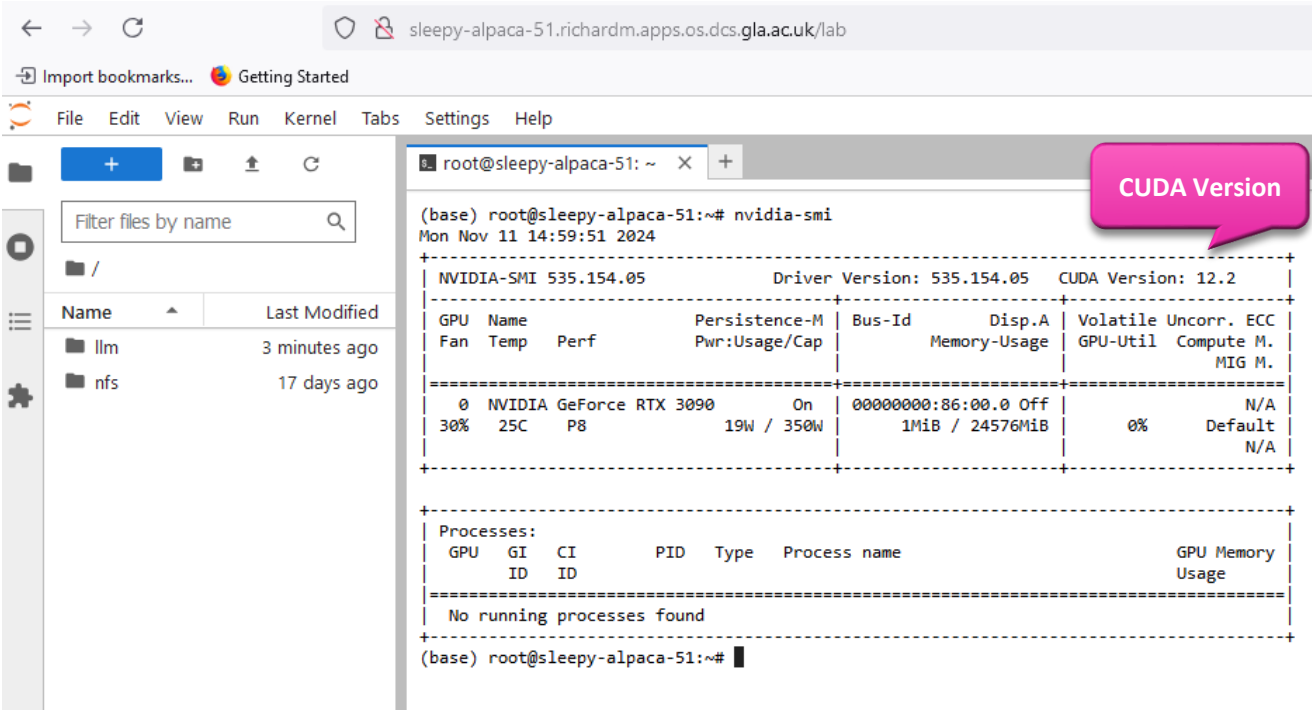
[Copy Pod](#)

Need help? Check out the [IDA Launcher Documentation](#).

Click here to open the notebook server in your browser

CUDA

- If you request a GPU as part of the resource request, then the container will have a recent version of the **CUDA** libraries installed, which is compatible with the cluster driver version
- If your account has a high-enough quota, you can request a container to have multiple GPUs in the advanced options during container configuration
 - Note that this is a difficult scheduling task, since all GPUs must be on the same physical host

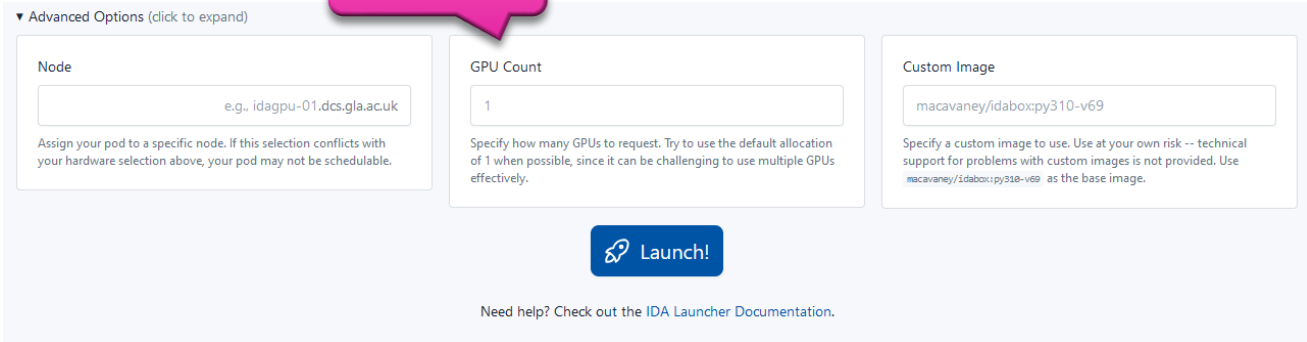


```
(base) root@sleepy-alpaca-51:~# nvidia-smi
Mon Nov 11 14:59:51 2024

+-----+
| NVIDIA-SMI 535.154.05                  Driver Version: 535.154.05      CUDA Version: 12.2               |
+-----+-----+
| GPU   Name                               Persistence-M   Bus-Id        Disp.A     Volatile Uncorr. ECC |
| Fan  Temp  Perf    Pwr:Usage/Cap       Memory-Usage   GPU-Util  Compute M. |
|=====|=====|
|  0  NVIDIA GeForce RTX 3090              On              00000000:86:00:0 Off      1MiB / 24576MiB      0%      Default |
| 30%   25C   P8             19W / 350W                               MIG M. |
+-----+-----+

+-----+
| Processes:                               GPU Memory |
|  GPU   GI    CI        PID   Type   Process name                      Usage |
|=====|=====|
| No running processes found |
+-----+

(base) root@sleepy-alpaca-51:~#
```



▼ Advanced Options (click to expand)

Node
e.g., idagpu-01.dcs.gla.ac.uk
Assign your pod to a specific node. If this selection conflicts with your hardware selection above, your pod may not be schedulable.

GPU Count
1
Specify how many GPUs to request. Try to use the default allocation of 1 when possible, since it can be challenging to use multiple GPUs effectively.

Custom Image
macavaney/idabocpy310-v69
Specify a custom image to use. Use at your own risk -- technical support for problems with custom images is not provided. Use macavaney/idabocpy310-v69 as the base image.

[Launch!](#)

Need help? Check out the [IDA Launcher Documentation](#).



Understanding Compute Usage

What we have covered so far should be sufficient to get you started with the cluster

However, as we are using a shared resource that does not have infinite capacity, it is important to discuss [usage policy](#), i.e. how we make sure people can get capacity when they need it

- The cluster operates on a [first-come-first-served](#) basis
- The maximum resources any one person can request is [capped via resource quota](#) on that person's [project](#)
- Workloads using the [IDA Container](#) have an [in-built time-out](#), where after a set period of CPU in-activity the pod will terminate
- The [cluster administration team](#) reserves the right to [kill workloads at any time](#) to redistribute capacity

Cluster users can request [additional capacity](#) if they are heading towards a deadline and need a burst of compute

- Email the cluster administrator for this

Usage Data is Open

- The cluster has a usage dashboard that everyone can use:

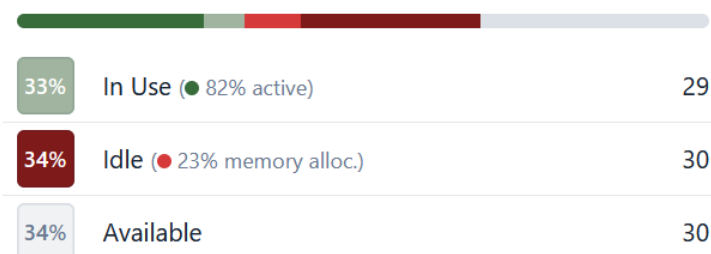
<http://usage.ida.dcs.gla.ac.uk/>



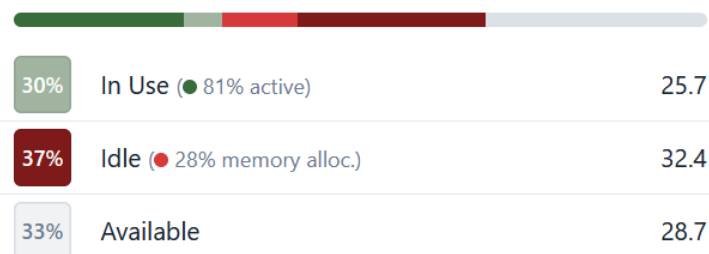
IDA Cluster Usage Dashboard

► This tool logs, parses, aggregates, and displays outputs from `/cluster.status`. There are several caveats, click to expand.

Mon 11 Nov 17:14



Average over Week



Legend

- ☐ **Available:** This GPU is ready to be used by anybody.
- ☒ **Idle:** Reserved but not in use and has no memory allocated.
- ☒ **Idle (Mem. Alloc.):** Idle but has GPU memory allocated.
- ☒ **In Use:** Reserved and has any GPU utilisation.
- ☒ **In Use (High Utilisation):** In use with high GPU utilisation.

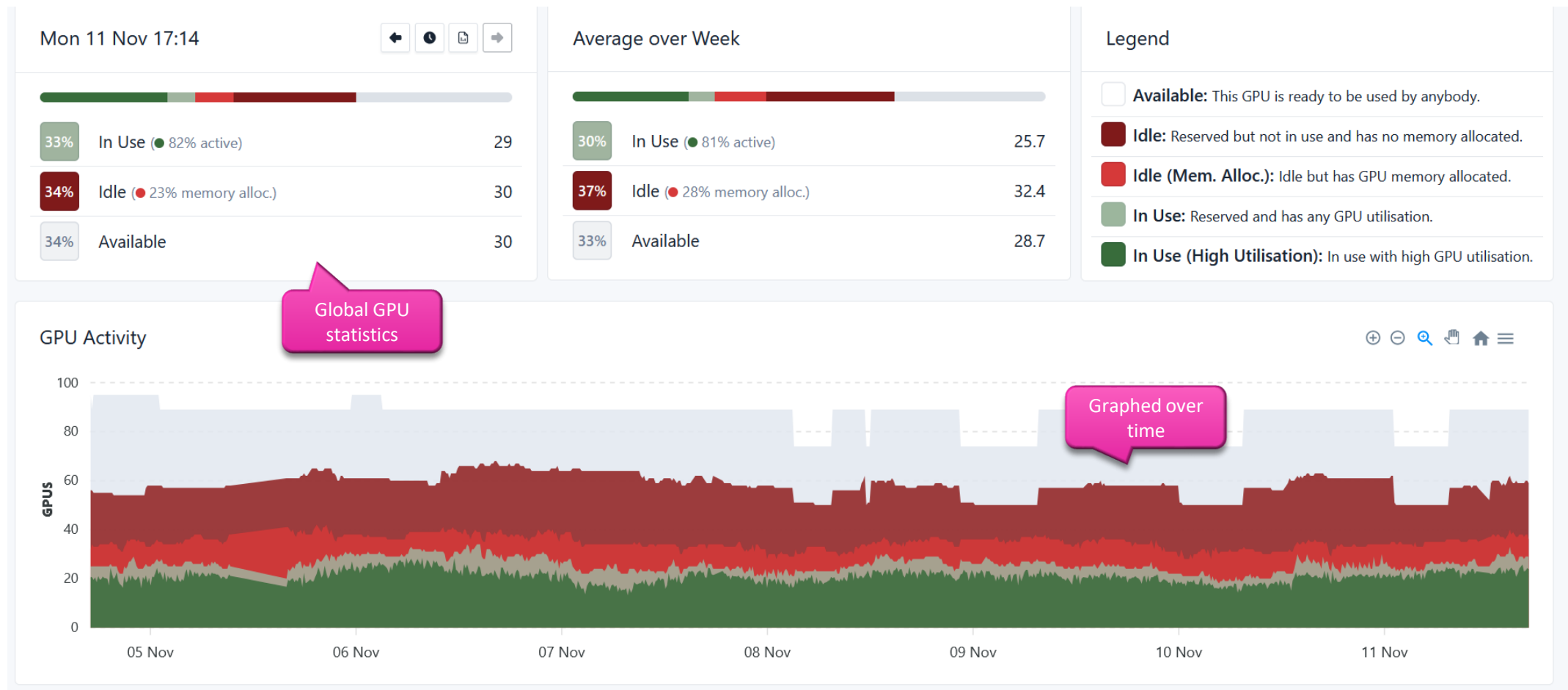
GPU Activity



100

Usage Dashboard

- The **bottleneck** in terms of capacity on the OKD4 cluster is **availability of GPUs**, and as such the dashboard is primarily reporting information about their availability and usage



Usage Dashboard

- The **bottleneck** in terms of capacity on the OKD4 cluster is **availability of GPUs**, and as such the dashboard is primarily reporting information about their availability and usage

GPU Activity Breakdown

Time

Filter...

Tue 05 Nov

Wed 06 Nov

Thu 07 Nov

Fri 08 Nov

Sat 09 Nov

Sun 10 Nov

Mon 11 Nov



GPU
Types

You can look
for empty
spaces to
find where
free GPUs
are

- The colour of the bar indicates usage:
 - White: Free
 - Red: Allocated but Idling
 - Green: Allocated and doing work

Usage Dashboard

- The **bottleneck** in terms of capacity on the OKD4 cluster is **availability of GPUs**, and as such the dashboard is primarily reporting information about their availability and usage

Users

USER	GPU HOURS RESERVED	GPU HOURS IDLE	GPU HOURS USED	% IN USE	% ACTIVITY	\$ IDLE	
l421andrew	701.6	1.3	700.3	100%	93%	\$1	
pgr22lubingzhi	506.7	42.4	464.3	92%	91%	\$0	
pgr23xinhao	506.1	86.5	419.6	83%	86%	\$58	
pgr23junchen	506.1	178.0	328.1	65%	88%	\$119	
pgr22zeyuan	506.1	68.3	437.8	87%	62%	\$46	
pgr22jinyuan	506.1	37.1	469.0	93%	90%	\$25	
pgr21zixuan	506.1	341.2	164.9	33%	84%	\$229	
ogechi	505.7	416.0	89.7	18%	2%	\$220	
2614994l	469.0	240.9	228.1	49%	92%	\$161	
pgr23kaiwen	461.5	301.3	160.2	35%	92%	\$202	
pgr23xizhang	355.7	18.3	337.4	95%	99%	\$12	
pgr23xuejun	342.3	205.8	136.6	40%	99%	\$138	
zijun	337.4	0.0	337.4	100%	100%	\$0	
pgr24shen	337.4	49.9	287.5	85%	94%	\$33	

- We also provide a summary of the recent usage for the main cluster consumers, along with how much money they are wasting by holding GPUs but not using them

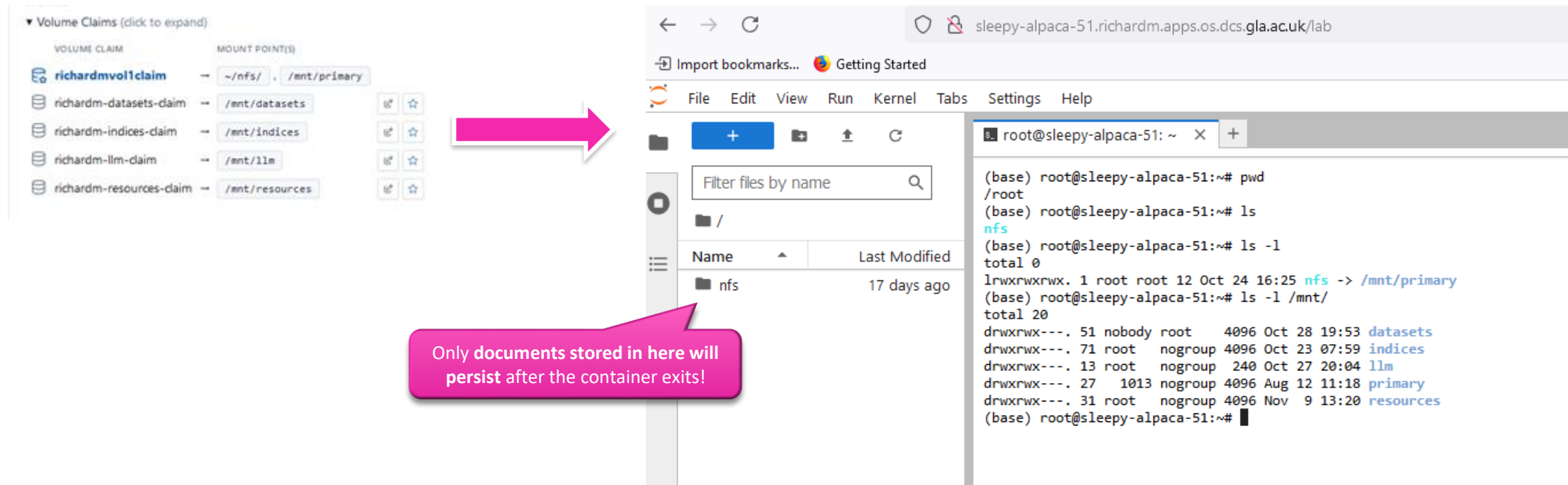
Only request a GPU if you are going to use it, and don't leave your Containers Idling!



Understanding Storage

Inside the Container

- As we noted earlier, by default the IDA Container will mount your storage to standard locations:



The image shows a Docker interface on the left and a file explorer on the right. A pink arrow points from the 'richardmvol1claim' volume to the 'nfs' directory in the file explorer. A pink callout bubble points to the 'nfs' directory with the text: 'Only documents stored in here will persist after the container exits!'.

Volume Claims (click to expand)

VOLUME CLAIM	MOUNT POINT(S)
richardmvol1claim	→ /nfs/ , /mnt/primary
richardm-datasets-claim	→ /mnt/datasets
richardm-indices-claim	→ /mnt/indices
richardm-llm-claim	→ /mnt/llm
richardm-resources-claim	→ /mnt/resources

File Explorer: sleepy-alpaca-51.richardm.apps.os.dcs.gla.ac.uk/lab

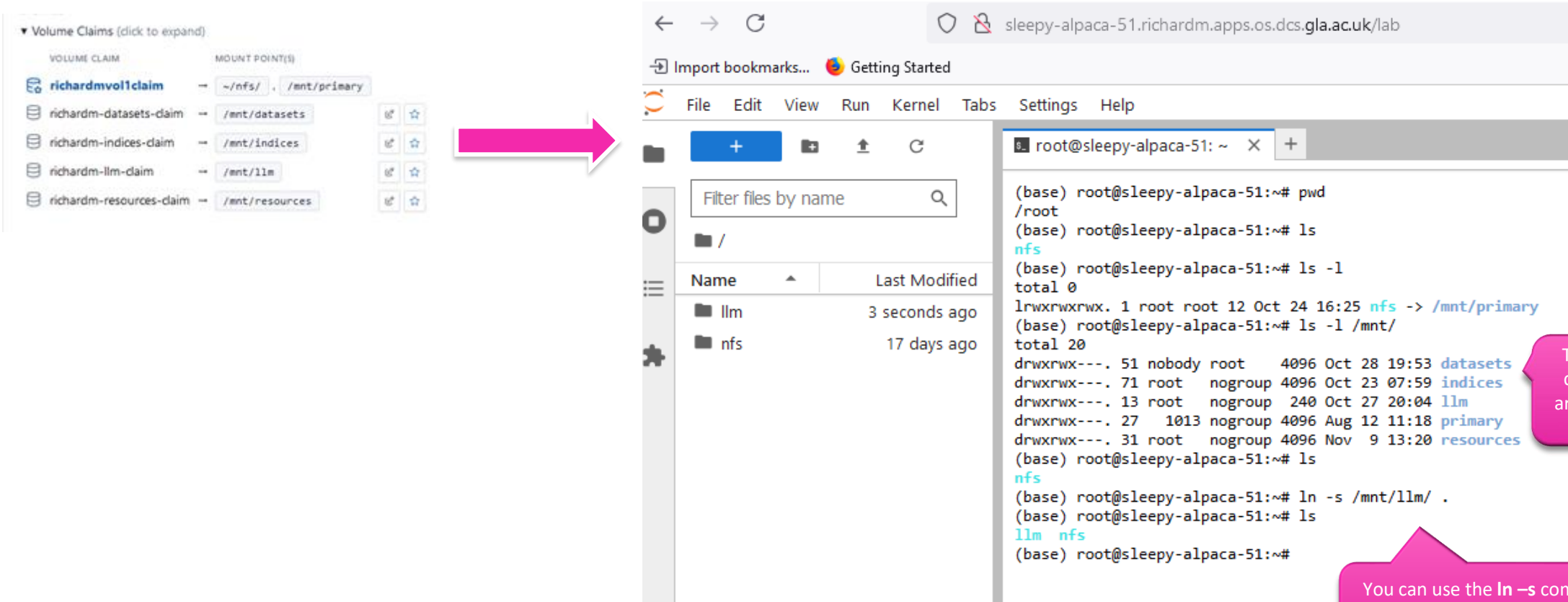
Filter files by name

Name	Last Modified
/	
nfs	17 days ago

```
(base) root@sleepy-alpaca-51:~# pwd
/root
(base) root@sleepy-alpaca-51:~# ls
nfs
(base) root@sleepy-alpaca-51:~# ls -l
total 0
lrwxrwxrwx. 1 root root 12 Oct 24 16:25 nfs -> /mnt/primary
(base) root@sleepy-alpaca-51:~# ls -l /mnt/
total 20
drwxrwx---. 51 nobody root 4096 Oct 28 19:53 datasets
drwxrwx---. 71 root nogroup 4096 Oct 23 07:59 indices
drwxrwx---. 13 root nogroup 240 Oct 27 20:04 llm
drwxrwx---. 27 1013 nogroup 4096 Aug 12 11:18 primary
drwxrwx---. 31 root nogroup 4096 Nov 9 13:20 resources
(base) root@sleepy-alpaca-51:~#
```

Inside the Container

- As we noted earlier, by default the IDA Container will mount your storage to standard locations:



The image shows the configuration of volume claims and the resulting file system structure inside a container. On the left, a table lists volume claims and their mount points. A pink arrow points from this table to a file explorer on the right, which shows the contents of the container's root directory. The file explorer shows a directory structure with 'llm' and 'nfs' folders. The terminal on the right shows the output of various commands, including 'pwd', 'ls', and 'ls -l /mnt/'. The terminal output shows that the directories 'datasets', 'indices', 'llm', 'primary', and 'resources' are mounted in the '/mnt/' directory. A pink callout box points to the '/mnt/' directory in the terminal output, stating: 'The shared directories are mounted in /mnt/'. Another pink callout box points to the 'ln -s' command in the terminal, stating: 'You can use the ln -s command to create a short-cut to these in your home directory'.

VOLUME CLAIM	MOUNT POINT(S)
richardmvol1claim	~/nfs/ , /mnt/primary
richardm-datasets-claim	/mnt/datasets
richardm-indices-claim	/mnt/indices
richardm-llm-claim	/mnt/llm
richardm-resources-claim	/mnt/resources

```
(base) root@sleepy-alpaca-51:~# pwd
/root
(base) root@sleepy-alpaca-51:~# ls
nfs
(base) root@sleepy-alpaca-51:~# ls -l
total 0
lrwxrwxrwx. 1 root root 12 Oct 24 16:25 nfs -> /mnt/primary
(base) root@sleepy-alpaca-51:~# ls -l /mnt/
total 20
drwxrwx---. 51 nobody root 4096 Oct 28 19:53 datasets
drwxrwx---. 71 root nogroup 4096 Oct 23 07:59 indices
drwxrwx---. 13 root nogroup 240 Oct 27 20:04 llm
drwxrwx---. 27 1013 nogroup 4096 Aug 12 11:18 primary
drwxrwx---. 31 root nogroup 4096 Nov 9 13:20 resources
(base) root@sleepy-alpaca-51:~# ls
nfs
(base) root@sleepy-alpaca-51:~# ln -s /mnt/llm/ .
(base) root@sleepy-alpaca-51:~# ls
llm nfs
(base) root@sleepy-alpaca-51:~#
```

The shared directories are mounted in /mnt/

You can use the ln -s command to create a short-cut to these in your home directory

Storage Capacity

Your **personal storage location**, which is mounted by default to

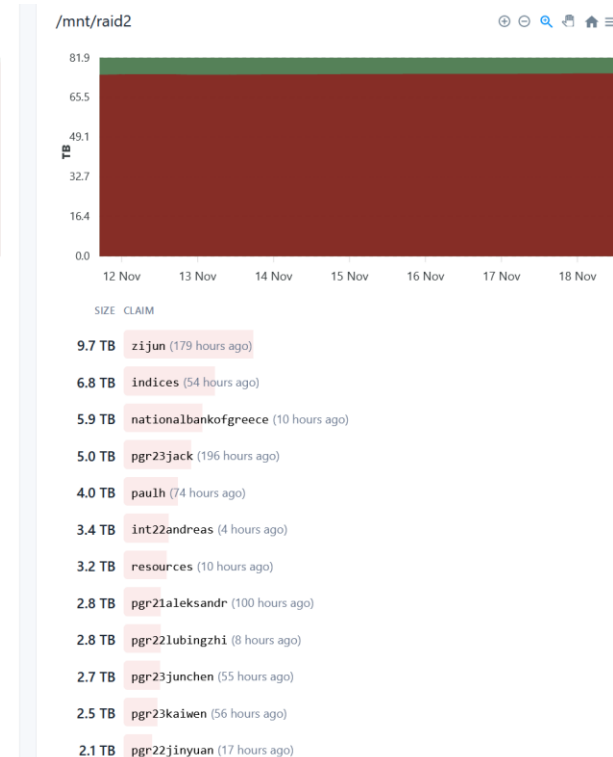
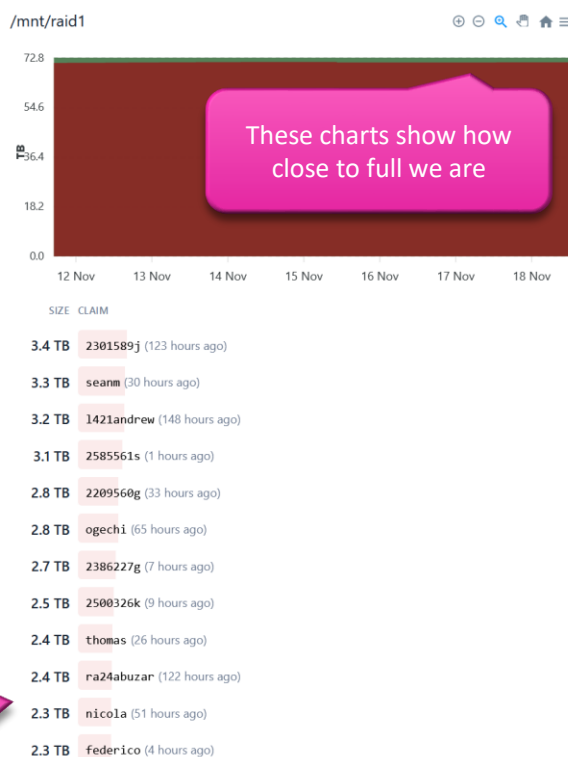
- ~/nfs/
- /mnt/primary

... **is not infinite** and the storage device are shared amongst a subset of the cluster users

This means that periodically storage **will run low**, and cluster users will be asked to delete or archive unused data

At the time of writing, there are two devices housing personal storage, denoted **raid1** and **raid2**

- The usage dashboard also shows storage consumption by user



You can find your storage consumption by looking for your identity in these lists

Summary

- Most users will start development from the **IDA Container**
 - A standard container that comes pre-loaded with common libraries, like a Linux machine in a box
 - Is designed to be compatible with our cluster
 - Users can customise it once launched
- Users start IDA Containers using the **Launcher App**
 - <https://launcher.apps.os.dcs.gla.ac.uk/>
 - Get up and running with a few clicks
 - Customisable resource requests
 - Mounts your personal storage (and shared storage)
- Resource usage can be monitored via the **Usage Dashboard**
 - <http://usage.ida.dcs.gla.ac.uk/>
 - How loaded is the cluster and what GPUs are free?
 - How much shared storage is available?

Questions?



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